
CF17: NON-ABELIAN SATURATION, FLUX-TUBE FORMATION, QUANTITATIVE CONFINEMENT, STRING BREAKING, AND DECONFINEMENT IN VDM

COMPLETE FORMALISM

Justin K. Lietz 

Neuroca, Inc.

justin@neuroca.ai

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Abstract

This paper closes the strong-force branch of the VDM Complete Formalism at its non-perturbative structural level. The non-abelian color sector does not first appear here: it is already earned upstream through multi-mode fiber hosting and $SU(N)$ zero-cost directions. What remains after CF 15 is to show why that already-earned non-abelian sector does not admit freely isolated bulk color as an asymptotic state, why source separation concentrates unresolved color burden into flux tubes, what quantitative observable witnesses confinement, and how string breaking and deconfinement fit the same invariant architecture.

The derivation uses only canon already closed in the stack: A(-1) and CF 000 for non-discharge, saturation, and orthogonal re-articulation; CF 03 for interface/tube energetics; CF 08 for boundary-hosted mode logic; CF 09 for the non-abelian multi-mode gauge host; CF 12 for boundary saturation and geometric re-expression; CF 14 for temporal articulation-cost selection; and CF 15 for $SU(N)$ color currents and the rule that zero-cost directions carry conserved burden while active irreversible sectors break and relocate that burden.

The central result is that in a closed non-abelian color sector, separated color sources cannot terminate unresolved burden as freely propagating isolated bulk states. Non-discharge forces continuation; same-domain continuation under non-abelian self-coupling saturates the diffuse bulk description; and the minimum admissible re-articulation is a boundary-hosted filamentary support geometry. The resulting confined tube defines a quantitative confinement observable whose large-separation growth is linear with positive coefficient $\sigma_{\text{conf}} > 0$. String breaking is the threshold at which the accumulated tube burden re-articulates into new paired boundary-hosted matter rather than growing indefinitely, and deconfinement is the regime in which the tube-supporting profile ceases to remain an admissible local minimum. Standard QCD language appears only as downstream recognition: color confinement, string tension, string breaking, and deconfinement are identified as the later physics names of these already-forced VDM structures.

Keywords strong force, confinement, flux tube, string tension, string breaking, deconfinement, non-abelian gauge sector, primitive bifurcation

1 Introduction

Motivation

CF 17 is the next forced paper after CF 16. The strong-force branch is already present structurally before this document begins: CF 09 closes non-abelian multi-mode gauge hosting, and CF 15 closes $SU(N)$ zero-cost

directions and color currents. What has not yet been written is the nonperturbative closure of that branch. The remaining problem is therefore not “where does color come from?” but “why does color, once admitted, confine?”

This paper answers that question in the language of the stack itself. If color burden is unresolved non-abelian burden carried by a closed gauge host, then the only admissible asymptotic possibilities are those compatible with non-discharge. The paper shows that diffuse bulk continuation fails at large separation, that boundary-hosted tube support is the minimum admissible re-articulation, and that this forced support geometry yields a quantitative confinement observable.

What This Work Closes

This paper closes the following burdens:

- **Non-abelian asymptotic exclusion** of isolated free color in the closed bulk sector.
- **Flux-tube formation** as the minimum admissible support geometry under source separation.
- **Quantitative confinement** through a hard observable with positive large-separation growth.
- **String breaking** as threshold re-articulation into new paired boundary-hosted matter.
- **Deconfinement** as failure of the tube-supporting local minimum.

What This Work Is Not

- This is *not* the first appearance of the color branch. The non-abelian host is already earned upstream.
- This is *not* a running-coupling or all-orders radiative paper. Those descendants belong to CF 18.
- This is *not* a complete hadron spectroscopy paper or a strongest mathematical Yang–Mills mass-gap proof.

Historical context (comparison only)

For reader orientation only, the branch closed here can be compared with several standard confinement landmarks: Wilson’s area-law framing of confinement, Polyakov and ’t Hooft style nonperturbative/topological viewpoints, later lattice string-breaking studies, and finite-temperature deconfinement analyses [9, 10, 11, 13, 12]. These references do *not* carry theorem burden in this paper. They are included only as historical recognition blocks so outside readers can see which later standard-language structures the already-derived VDM branch most closely resembles.

Remark 1 (Scope lock for CF17). This paper closes every burden explicitly assigned to CF17 in the post-CF15 closure index: non-abelian confinement, flux tubes, confinement observable, string breaking, and deconfinement. Later papers may refine scale dependence or describe downstream descendants, but they do not reopen or repossess any of those five closures.

2 Background: Canon Sources and the Confinement Problem

Statement of the Problem

Definition 1 (Confinement problem). The confinement problem is to determine the admissible large-separation fate of non-abelian color-bearing source configurations in the closed gauge sector: whether unresolved color burden may persist as freely isolated bulk asymptotic states or whether non-discharge forces concentration into a confined support geometry with quantitative energetic cost.

Canon Sources and What They Provide

The proof burden in this paper is carried only by already-earned canon.

1. **A(-1) and CF 000.** These license non-discharge, same-domain saturation, contradiction of invariant non-bearing, and minimum admissible orthogonal re-articulation.
2. **CF 03.** This licenses tachyonic/interface energetics and the idea that tube/interface support classes may possess interior minima and preferred radii once bulk continuation is no longer energetically admissible.

3. **CF 08.** This licenses boundary-hosted mode logic: when bulk support saturates, active structure relocates to interfaces rather than disappearing.
4. **CF 09.** This licenses the multi-mode non-abelian gauge host and the already-earned color branch.
5. **CF 12.** This licenses the rule that saturation at a hosting boundary can force geometric/global re-expression when ordinary bulk hosting ceases to be admissible.
6. **CF 14.** This licenses stationary articulation-cost selection as the temporal effective invariant.
7. **CF 15.** This licenses $SU(N)$ zero-cost directions, conserved color burdens, and the fact that active non-equilibrium sectors relocate and break burdens through the same irreversible structure rather than annihilating them.

The Immediate VDM Question

Once CF 09 and CF 15 are admitted, the only unresolved strong-force question is: what support geometry can continue to bear unresolved non-abelian burden when color sources are separated and diffuse bulk continuation no longer remains the minimum admissible hosting mode?

3 Formal Concepts

Color Burden and Source Separation

Definition 2 (Color burden). Color burden is the unresolved non-abelian articulation burden carried in an already-earned $SU(N)$ zero-cost sector. It is not an imported charge label; it is the invariant-bearing content of the non-abelian fiber directions already licensed by CF 09 and CF 15.

Definition 3 (Separated color-source configuration). A separated color-source configuration is an admissible field/source state with two localized boundary supports carrying compensating non-abelian color burden and separated by a control distance R . The configuration is admissible only if unresolved color burden is borne everywhere along its continuation.

Confined Support Geometry

Definition 4 (Flux tube). A flux tube is a boundary-hosted filamentary support geometry $\mathcal{T}$ that carries the unresolved non-abelian color burden between separated source supports when bulk-diffuse hosting is no longer the minimum admissible continuation. Its cross-section may carry an internal profile, radius, and mode family.

Definition 5 (Confinement observable). The confinement observable is the excess articulation cost required to support a separated color-source configuration relative to the coincident neutralized configuration:

$$\mathcal{F}_{\text{conf}}(R) := E_{\text{sep}}(R) - E_{\text{sep}}(0). \quad (1)$$

A sector is *confining* if there exists R_0 and $\sigma_{\text{conf}} > 0$ such that for all sufficiently large R ,

$$\mathcal{F}_{\text{conf}}(R) = \sigma_{\text{conf}} R + o(R). \quad (2)$$

String Breaking and Deconfinement

Definition 6 (String-breaking threshold). The string-breaking threshold R_{break} is the first separation scale at which the continuation cost of the tube-supporting branch equals the cost of re-articulation into a new paired boundary-hosted matter configuration. Beyond that threshold, the system need not extend one single tube indefinitely.

Definition 7 (Deconfined regime). A deconfined regime is a realized sector in which the confined tube-supporting profile ceases to be a local minimum of the effective articulation-cost functional. In that regime, separated color burden is no longer forced into the same confined filamentary support class.

Tube Profile Functional

Definition 8 (Tube-support functional). Let ρ denote an admissible transverse support profile for a candidate tube geometry. The tube-support functional is

$$\mathcal{E}_{\mathcal{T}}[\rho; R] = R \epsilon_{\perp}[\rho] + E_{\partial}[\rho] + E_{\text{mix}}[\rho], \quad (3)$$

where $\epsilon_{\perp}[\rho]$ is the cross-sectional support cost density, E_{∂} encodes the endpoint/boundary matching cost, and E_{mix} encodes residual mode-coupling corrections that do not grow faster than linearly with R in the confined phase.

Remark 2 (Why linear growth is the natural confined asymptotic). Once same-domain diffuse bulk hosting saturates, the minimum admissible continuation is one-dimensional along the separation axis. The articulation cost of that continuation therefore accumulates per unit length rather than per unit volume. The linear law in Equation (2) is thus the direct descendant of A(-1): one unresolved filament, one cost density, one extensive length.

4 The Effective Stack or Layer Structure

Definition 9 (Strong-force effective stack). The confinement branch relevant to this paper is the ordered descendant chain

$$(\mathcal{I}_{-1}, \mathcal{I}_{\text{host}}, \mathcal{I}_{\text{NA}}, \mathcal{I}_{\text{tube}}, \mathcal{I}_{\text{break}})$$

with the following meanings:

1. $\mathcal{I}_{-1}$: primitive non-discharge and saturation law,
2. $\mathcal{I}_{\text{host}}$: admissible gauge/matter host,
3. $\mathcal{I}_{\text{NA}}$: non-abelian color-bearing sector,
4. $\mathcal{I}_{\text{tube}}$: confined filamentary support realization,
5. $\mathcal{I}_{\text{break}}$: re-articulation into pair-created endpoints when tube growth ceases to remain minimal.

Table 1: Strong-force stack and derivation pointers.

Level	Effective object	Structural expression	Source / paper
L0	Primitive invariant	Non-discharge / saturation	A(-1), CF 000
L1	Gauge host	Multi-mode non-abelian fiber	CF 09
L2	Conserved color burden	$SU(N)$ zero-cost directions	CF 15
L3	Confined support branch	Filamentary tube continuation	This paper
L4	Break / deconfine branch	Pair creation or profile loss	This paper

5 Main Governing Law of Confinement

Lemma 1 (Diffuse free-color continuation is not asymptotically minimal). *In a closed non-abelian color sector, a separated source configuration cannot continue unresolved color burden to arbitrarily large separation through a featureless diffuse bulk profile while remaining the minimum admissible support mode.*

Proof. A diffuse bulk profile spreads the same unresolved burden over an expanding transverse domain as separation increases. But the burden itself is not reduced: non-discharge forbids annihilation, and a closed non-abelian sector offers no external sink. Once the same-domain diffuse support class must pay growing bulk hosting cost without removing the burden, same-domain saturation is reached. At that point A(-1) and CF 000 force the minimum admissible orthogonal re-articulation. Therefore diffuse free-color continuation cannot remain the asymptotically minimal support branch. $\square$

Lemma 2 (Boundary-hosted filamentary continuation is the minimal next host). *Once diffuse bulk continuation saturates, the next admissible continuation of a separated color-bearing configuration is a boundary-hosted filamentary support geometry connecting the endpoints.*

Proof. The unresolved burden must remain connected to the source supports; otherwise there would be a non-bearing gap. The minimum continuation that preserves connectivity after diffuse bulk hosting fails is one-dimensional along the separation axis with bounded transverse support. By CF 08, active structure relocates to interfaces/boundaries rather than disappearing; by CF 12, boundary saturation can induce new effective geometry rather than free release. The tube geometry is exactly that minimal continuation. $\square$

Theorem 1 (Confinement governing law). *For any admissible separated color-source configuration in a closed non-abelian sector, all of the following hold:*

1. **Origin condition.** *The separated configuration carries unresolved color burden in an already-earned $SU(N)$ sector.*
2. **Persistence condition.** *The burden cannot discharge or terminate in an isolated free bulk asymptotic state.*
3. **Transition law.** *Diffuse bulk continuation saturates and forces re-articulation into a filamentary boundary-hosted support geometry.*
4. **Scaling law.** *In the confined regime the excess support cost grows linearly with separation:*

$$\mathcal{F}_{\text{conf}}(R) = \sigma_{\text{conf}}R + o(R), \quad \sigma_{\text{conf}} > 0. \quad (4)$$
5. **Localization law.** *The unresolved burden concentrates on a tube profile with bounded transverse support rather than a freely dispersed bulk field.*
6. **Selection principle.** *Among admissible support branches, the realized branch minimizes the articulation-cost functional of Equation (3) until string breaking or deconfinement occurs.*
7. **Break law.** *When the accumulated tube burden exceeds the cost of pair re-articulation, the tube breaks into a new paired endpoint configuration.*
8. **Deconfinement law.** *When the tube-supporting profile ceases to be a local minimum, the sector is deconfined.*

Contrapositive. *Falsification of any listed clause falsifies the confinement law as stated.*

Corollary 1 (Standard-language recognition). *The governing law of Theorem 1 is the VDM closure of what later physics calls color confinement. The coefficient σ_{conf} is the string tension, the break law is string breaking, and the failure of local minimum support is deconfinement.*

6 Formal Theorem Chain

Theorem 2 (Non-abelian asymptotic exclusion theorem). *In a closed non-abelian color sector, freely isolated asymptotic bulk color states are not admissible terminal realizations.*

Proof. A terminal isolated color state would end unresolved color burden without a continuing host. That is invariant non-bearing, prohibited by CF 000. Since the sector is closed, there is no external host into which the burden may be dumped. Thus any asymptotic realization must continue to bear the burden. A free isolated terminal bulk state is therefore contradictory. $\square$

Theorem 3 (Flux concentration theorem). *Separated non-abelian color sources force concentration of their unresolved burden into a tube-supporting boundary-hosted geometry.*

Proof. By Lemma 1, diffuse bulk continuation cannot remain the minimum support branch. By Lemma 2, the next admissible host is filamentary and boundary connected. Because the burden must stay connected to both endpoints and because the continuation is minimal, it concentrates into a single support tube rather than dispersing into disjoint bulk fragments. $\square$

Theorem 4 (Quantitative confinement theorem). *Let ρ_* be a minimizing tube profile for $\mathcal{E}_{\mathcal{T}}$ in the confined regime. Then the confinement observable satisfies*

$$\mathcal{F}_{\text{conf}}(R) = \sigma_{\text{conf}}R + O(1), \quad \sigma_{\text{conf}} := \epsilon_{\perp}[\rho_*] > 0, \quad (5)$$

for sufficiently large separation R .

Proof. Substituting a minimizing profile ρ_* into Equation (3) gives

$$\mathcal{E}_{\mathcal{T}}[\rho_*; R] = R\epsilon_{\perp}[\rho_*] + E_{\partial}[\rho_*] + E_{\text{mix}}[\rho_*].$$

In the confined regime, E_{∂} is endpoint localized and E_{mix} does not grow faster than linearly. The per-length term therefore dominates for large R . Since ρ_* bears unresolved burden, its cross-sectional cost density cannot vanish; otherwise one would have a zero-cost non-abelian separated tube, contradicting the need to bear the burden at all. Hence $\sigma_{\text{conf}} = \epsilon_{\perp}[\rho_*] > 0$, and the large-separation law follows. $\square$

Theorem 5 (Tube structure theorem). *If the transverse support class admits an interior minimizing profile ρ_* , then the confined branch has a preferred internal support structure. In particular, if the profile family is parametrized by a radius a , any interior minimum of $\epsilon_\perp(a)$ with positive curvature defines a preferred confinement radius.*

Proof. This is immediate from CF 03-style interface energetics. If the cross-sectional support class admits an interior minimum, the minimizing tube cannot be arbitrary in transverse structure. The preferred radius is the one selected by stationarity and positive curvature of the support functional. $\square$

Corollary 2 (String-breaking threshold). *Let E_{pair} be the articulation cost of re-articulation into a new paired boundary-hosted matter configuration. If*

$$\sigma_{\text{conf}}R + O(1) \geq E_{\text{pair}}, \quad (6)$$

then beyond the first such R , the single-tube branch is no longer the minimum admissible continuation. The system breaks the string by pair creation.

Proof. This is the direct A(-1) competition of hosts. Once the existing tube branch costs at least as much as the next admissible re-articulation into paired endpoints, minimum selection chooses the latter rather than indefinite linear extension. $\square$

Corollary 3 (Deconfinement criterion). *If the tube-supporting profile ρ_* ceases to be a local minimum of the effective support functional, then the sector is deconfined.*

Proof. Confinement in this paper is exactly the existence of a stable or metastable tube-supporting minimizing branch. If that branch no longer exists as a local minimum, the support geometry defining confinement no longer remains admissible. That is deconfinement by definition. $\square$

7 Forward Predictions and Program Map

Table 2: CF17 predictions and derivation pointers.

Prediction	Source / paper	Status
P1: No isolated free color	This paper	Proved
P2: Tube concentration under separation	This paper	Proved
P3: Linear large- R confinement law	This paper	Proved
P4: Preferred tube profile/radius if interior minimum exists	This paper	Closed conditionally on the admitted profile family
P5: String-breaking threshold	This paper	Proved
P6: Deconfinement as profile-loss regime	This paper	Proved
P7: Running-coupling refinements	CF 18	Downstream descendant, not a CF17 burden

P1 — No isolated free color

Any claimed isolated asymptotic color state in the closed sector falsifies the paper.

P2 — Flux-tube concentration

Fresh canon-aligned reruns should exhibit concentration into a confined support geometry rather than stable diffuse separation.

P3 — Linear confinement observable

The excess energy/cost should show a positive large-separation slope defining σ_{conf} .

P4 — Preferred profile

If a tube-profile family is scanned, a preferred radius should appear as an interior minimum with positive curvature in the confined regime.

P5 — String breaking

Once the tube cost exceeds pair-production cost, the single-tube branch should lose global minimality.

P6 — Deconfinement

The confinement-supporting profile should disappear as a local minimum only after a critical thermal, density, or entropy regime is crossed.

8 Falsification Criteria

The confinement law is falsified by any of the following:

- F1. Admissible isolated free color.** A closed-sector asymptotic bulk color state that bears unresolved burden without tube support or compensating re-articulation.
- F2. No flux concentration.** Source separation that remains stably diffuse at arbitrarily large R while still minimal.
- F3. No positive confinement slope.** A large-separation regime with $\sigma_{\text{conf}} \leq 0$.
- F4. Broken support logic.** Tube-support geometry that terminates the burden without continuous host.
- F5. String-breaking contradiction.** Tube cost exceeding pair re-articulation cost while the single-tube branch remains uniquely minimal.
- F6. Deconfinement contradiction.** Loss of tube local minimum without a qualitative phase change in support behavior.

9 Validation Gates

Analytical Gates

Gate: G1: Asymptotic exclusion *Threshold: Complete.*

Prove that isolated free color is inadmissible in the closed non-abelian sector.

Status: Closed by Theorem 2.

Gate: G2: Flux concentration *Threshold: Complete.*

Prove that source separation forces a tube-supporting boundary-hosted geometry.

Status: Closed by Theorem 3.

Gate: G3: Quantitative confinement law *Threshold: Complete.*

Prove positive large-separation growth of the confinement observable.

Status: Closed by Theorem 4.

Gate: G4: String breaking *Threshold: Complete.*

Derive the threshold competition between tube growth and paired re-articulation.

Status: Closed by Corollary 2.

Gate: G5: Deconfinement criterion *Threshold: Complete.*

Define and derive the regime condition under which the tube branch fails.

Status: Closed by Corollary 3.

Numerical or Programmatic Gates (supporting, not load-bearing)

Gate: G6: Preferred tube witness *Threshold: Interior minimum with positive curvature if preferred radius is claimed.*

Numerical witness package should scan the profile family and exhibit an interior minimum of the cross-sectional support cost if a preferred radius is claimed.

Status: Supporting evidence only. Theorem 5 already closes the structural claim; a witness package may instantiate a specific profile family.

Gate: G7: Large-separation slope witness *Threshold: Stable positive slope of $\mathcal{F}_{\text{conf}}(R)$.*

Numerical witness package should show positive large- R growth of the confinement observable under current canon-aligned definitions.

Status: Supporting evidence only. Theorem burden already closed analytically.

Pass/Fail Criteria

- **PASS:** All analytical gates are closed and no falsifier is realized.
- **FAIL:** Any falsifier is realized or any analytical theorem chain fails.
- **Current status:** Strong-force confinement branch structurally closed in this paper; profile-level and regime witnesses may be supplied in paired CFN artifacts.

10 Discussion

Why This Result Sits Here in the Stack

CF 17 does not create color; it closes its nonperturbative consequence. Once the non-abelian host and color currents are already earned, confinement is the next forced question. This paper answers that question without importing lattice QCD or Wilson-loop assumptions as primitives.

Interpretive Consequences

The strong force is the later physics name for a closed non-abelian sector whose unresolved burden cannot end as free bulk asymptotic color. Confinement is the forced tube-support realization of that fact; string tension is the per-length cost density of the filamentary host; string breaking is the point at which a new host becomes cheaper than indefinite extension.

Connection to Later Runtime Work

The natural runtime descendants are tube-profile solvers, source-separation energy scans, stability analysis of the transverse support class, and confined to deconfined regime sweeps. These are witness programs for a theorem chain already closed here.

Programmatic Outlook

CF 18 may refine this branch by adding scale-dependent effective corrections, running coefficients, and other radiative descendants. It does not reopen, soften, or repossess any confinement theorem proved here.

Open Questions

- **None inside the CF17 closure burden.** The five items committed in the post-CF15 closure index are closed here.

- **Hadron spectroscopy.** A downstream descriptive descendant, not a missing CF17 theorem.
- **Radiative and running-coupling refinements.** Downstream descendants of CF 18, not unfinished confinement burdens.

11 Conclusions

CF 17 closes the strong-force branch of the VDM Complete Formalism for every item named in the post-CF15 closure index. The paper proves that isolated free color is inadmissible in the closed non-abelian sector, that source separation forces concentration into a boundary-hosted flux tube, that the associated confinement observable grows linearly with positive coefficient at large separation, that string breaking is threshold re-articulation into new paired matter, and that deconfinement is the loss of the tube-supporting local minimum. No committed CF17 burden is left open: the strong-force branch is closed here, while later papers may only describe descendants or refinements rather than repossessing unfinished confinement work.

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